AEGIS-RF — Icicle Kit Board-Document Audit

Purpose: translate the five supplied Microchip board documents into a defensible hardware plan for the PolarFire FPGA Design Contest Track 2 proposal.

Documents reviewed

Document	Revision / ID	Used for
1. PolarFire SoC Icicle Kit User Guide	DS60001679E, 2026 edition	Board identity, memory, I/O banks, expansion connectors, PAC1934 channels, LEDs/switches, 50 MHz fabric oscillator.
2. PolarFire SoC Icicle Kit QuickStart Guide	DS50003930A, 2025	254K logic elements, board power/USB-UART bring-up, Linux serial access and jumper baseline.
3. PolarFire SoC Icicle Kit Brochure	DS00003611A, 2020	Five-core RISC-V summary, 2 GB LPDDR4 ×32, SPI flash, eMMC/SD, dual Gigabit Ethernet, PCIe and expansion interfaces.
4. Icicle Kit Rev. 1.0 Schematics	DVP-102-000536-001, 2020	Actual board wiring, bank assignments, LPDDR4, clock, power and peripheral connectivity; board revision must match the physical kit.
5. PolarFire SoC FPGA Board Design Guidelines	DS60001681H, 2026 edition	Power integrity, I/O bank voltages, clock/reset, DDR and high-speed interface constraints.

Decisions carried into the full proposal

- **Use fabric for the deterministic workload:** Fixed FFT (only after the exact CoreFFT IP/version and licensing are confirmed), spectral power, full-band low-cost candidate awareness, and candidate-window detail/CFAR. Do not claim runtime FFT-length switching.
- **Use RISC-V/Linux for supervisory work:** Confidence/state machine, occupancy history, hysteresis, experiment orchestration, logging and Ethernet dashboard. Verify the supported BSP and register path.
- **Use vendor reference design before custom integration:** Boot the supported image and run the official AXI4_STREAM_DEMO/CoreAXI4DMAController example before modifying it. Pin commit/tag and tool version.
- **Treat LPDDR4 access as an integration gate:** The board documents 2 GB LPDDR4 ×32 on the MSS interface. Fabric access and DMA buffers require proof of the actual interconnect, address mapping, Linux API and cache coherency.
- **Measure only what the board exposes:** PAC1934 channels are documented as VDD, VDD25, VDDA25 and VDDA over MSS I2C. These are not automatically total-board, LPDDR4 or isolated fabric power.
- **Use board I/O where it improves the demo:** Two Gigabit Ethernet ports for result/experiment telemetry; four user LEDs and four buttons for visible mode/status; four UART channels for boot/debug. Validate muxing and pin constraints.
- **Keep optional interfaces out of the MVP:** PCIe Gen 2/four SERDES lanes, Raspberry Pi header and mikroBUS are possible extensions but not needed for the first working result. No RF ADC/SDR is included in the standard kit contents.

- **Use reproducible synthetic input first:** MATLAB-labelled signed complex I/Q is the primary input; recorded SDR files are an optional later extension and require a documented transfer route.

Evidence gates before claiming feasibility

1. Confirm exact physical board revision and device part/suffix; match the reference design's Libero release and constraints.
2. Boot the unmodified vendor design and capture UART/Linux logs.
3. Build and pass the official AXI4 streaming/DMA demo before inserting custom DSP.
4. Confirm CoreFFT availability, fixed transform length, stream handshake, data width, scaling, latency, license and resource cost.
5. Show a verified fabric-to-software data path; test address map, cache coherency, buffer ownership, interrupt handling and overflow recovery.
6. Produce synthesis/place-and-route reports listing LE, math-block, LSRAM, clock and timing usage. The supplied board documents do not justify invented utilisation numbers.
7. Calibrate PAC1934 channel mapping/update rate and synchronise readings with workload; use an external calibrated meter for whole-board energy.
8. Run fixed-high, fixed-low and adaptive configurations on the same labelled held-out I/Q scenarios at comparable false-alarm operating points.

Official contest-fit note

The supplied official rules weight proposal-stage Technical Feasibility at 35%, Innovation & Creativity at 25%, Problem Statement & Solution Clarity at 20%, and Potential Impact at 20%. The revised proposal maps the design to these criteria, but a high score cannot be guaranteed without implementation evidence. The same rules document says the final-stage project documentation must be no more than 10 pages; this full engineering proposal is a working specification and should not be submitted as the final report unchanged.

Source links

- Official Icicle Kit reference design repository: https://github.com/polarfire-soc/icicle-kit-reference-design
- Official Microchip PolarFire SoC product documentation: https://www.microchip.com/en-us/products/fpgas-and-plds/system-on-chip-fpgas/polarfire-soc-fpgas